

TABLE II
COST OF SEMICONDUCTOR COMPONENTS USED IN MEDICAL ELECTRONICS*

	ECG	EEG	Bedside monitor	Cath lab	Mammography	Digital hearing aid
Cost of the product	₹ 25,000 - ₹ 125,000 (US\$ 530 - US\$ 2669)	₹ 125,000 (US\$ 2669)	Upto ₹ 200,000 (US\$ 4255)	₹ 10,000,000 (US\$ 0.2mn)	₹ 1,000,000 - ₹ 1,500,000 (US\$ 21,000 - US\$ 31,915)	₹ 10,000 - ₹ 40,000 (US\$ 210 - US\$ 850)
Per cent cost of electronic components**	25%	25%	35%	25%	35%	50%
Per cent of imports	80 to 90 per cent of the electronic components are imported					
Key import locations	The USA, Germany, France, China, Singapore, Hong Kong and Korea					

*Indicative; **Includes semiconductor cost; Source: IESA

healthcare, and the need of the hour is for India to manufacture such equipment, as part of Make in India programme.

The National Policy on Electronics has medical electronics as one of the thrust areas and provides for financial incentives for medical electronics manufacturing not only for new units but also for units relocating from outside India. The Drugs and Cosmetics (Amendment) Bill, 2013, is now considering recognising medical devices as a separate category from pharmaceutical products in the regulatory structure. A strong framework that provides for medical devices standards in India is imperative to boost domestic manufacturing, bring in reliability and better time to market new technologies in medical electronics.

Recently, a memorandum of understanding was signed between Society for Applied Microwave Electronics Engineering & Research (SAMEER) and the government of Andhra Pradesh, on conducting joint and comprehensive assessment of health technologies as identified, starting with dual-energy linear accelerators for cancer care development. SAMEER is an R&D laboratory of the Ministry of Electronics and Information Technology, government of India, mandated to contribute towards the growth of science and technology of radio frequency/microwave electronics, electromag-

netic interference/compatibility and allied areas through intensive research, design, development and industrial applications.

Moving forward

The Indian medical electronics industry is till date highly under-invested and import-driven. According to a Boston Consulting Group report, India imports an estimated 70 per cent of medical devices. Lack of maturity in this sector with respect to penetration, accessibility, affordability and awareness possess major constraint for the growth of this sector in India. Moreover, nascent regulatory environment, no distinct status of the industry, need for quality benchmark at par with global standards and high capex requirement also slow down the growth of the indigenous manufacturing ecosystem.

However, an FICCI report cited examples showing a slow but gradual move towards Make in India in this sector. Sector players have undertaken endeavours to start manufacturing innovative products in India, which are suitable for domestic as well as overseas markets. For example, Transasia Biomedicals has developed in-vitro diagnostic equipment through its R&D base in Mumbai, Roche Diagnostics has developed a screening device for cardio-vascular diseases, suitable for use in rural settings, and GE Healthcare has developed a low-cost ECG

machine and a low-cost ultrasound machine for the Indian market.

Philips Healthcare has been using its recent acquisitions in India to develop and launch a low-cost cath lab for the Indian market.

The same report also mentions global investment in this sector. Global companies have started considering India as a manufacturing base by either setting up facilities of their own or by acquiring domestic manufacturers. Some examples include 3M's manufacturing plant in Pune, Becton Dickinson's manufacturing facility in Haryana, Hollister's setting up manufacturing facility in India and Philips Medical Systems' acquisition of Medtronics and Alpha X-Ray Technologies.

Medical technology parks have been proposed by the government in addition to existing parks to encourage domestic manufacturing of medical equipment. Inflow of FDI will spur both R&D and manufacturing, in turn, increasing the efficiency and effectiveness of medical electronics products.

Advancement of medical electronics product quality and associated successful diagnostic rates are expected to create a spurt in adoption. India as an emerging market continues to be relevant in the global medical electronics space, not just from a market point of view, but also as a global hub for innovation and manufacturing. **EFY**